

# Power functions



1

a Odd

b Even

c Even

d Odd

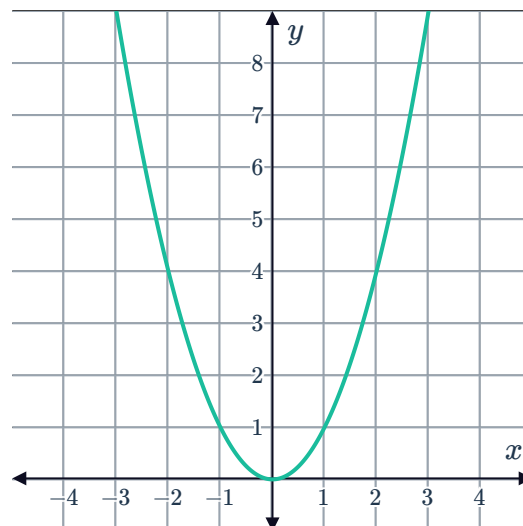
e Odd

2

a

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 9  | 4  | 1  | 0 | 1 | 4 | 9 |

b



c No

e 0

d  $x = 0$

f 2

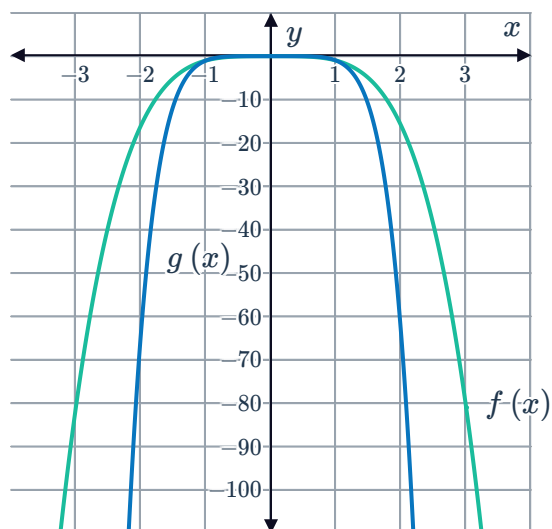
3

a Fall

b Rise

4

a

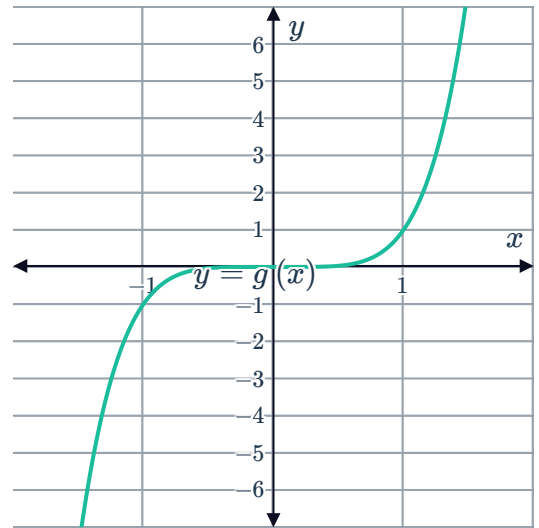
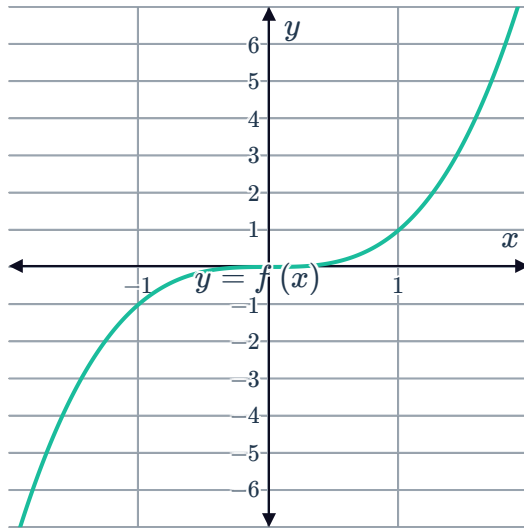


b  $g(x) = -x^6$

5



a

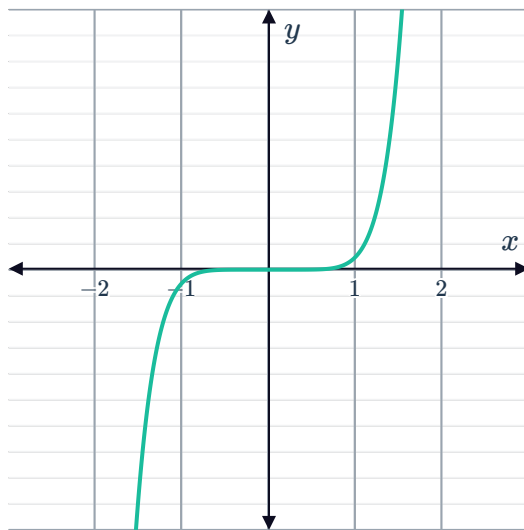


- b  $y = x^7$  would have the flattest gradient close to the origin, and the steepest gradient away from the origin.

6

- a They approach positive infinity.

c



- b They approach negative infinity.

d

